

Vectorial continued fractions and an algebraic construction of effective Hamiltonians

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In the harmonic oscillator basis, the effective Hamiltonians are constructed for the polynomial interactions $V(r) = \sum_{i=1}^t g_i r^{2i}$ and $t \leq 4$. Their form (factorized in terms of the vectorial continued fractions) is simpler in a semirelativistic ($c < \infty$) reformulation. Its degeneracy in the $c \rightarrow \infty$ limit (confluence of the physical and unphysical solutions) is removed by solving a coupled set of the nonlinear algebraic equations in REDUCE.

I. INTRODUCTION

The general anharmonic oscillator

$$\left(-\frac{d}{dr^2} + \frac{l(l+1)}{r^2} + g_1 r^2 + g_2 r^4 + \cdots + g_t r^{2t} \right) \psi(r) = E \psi(r), \quad g_i > 0, \quad l = (-1, 0, 1, \dots), \quad (1.1)$$

is a phenomenological as well as methodical laboratory in quantum mechanics.¹ Its use ranges from a purely nonrelativistic description of confinement² up to an analysis of the perturbative and nonperturbative aspects of the quantum field theory.³ In the former case, a maximal numerical efficiency is usually required. The present study of Eq. (1.1) is intended to complement the latter [e.g., Wentzel-Kramers-Brillouin (WKB) or perturbative⁴] type of applications, where an emphasis is laid upon a global and analytic insight into the solutions.

We shall start from the recurrent (matrix continued fractional, MCF,⁵ and vectorially continued fractional, VCF⁶) solutions of equations of the type (1.1). In brief, recalling accelerations of their convergence as achieved by a systematic subtraction of the so-called fixed-point (FP) approximants,⁷ we shall deliver the explicit nonperturbative approximate forms of the corresponding solvable (finite-dimensional) effective Hamiltonians⁸ for $t = 2, 3$, and 4.

First, we describe the recurrent factorization method in its entirely universal form in Sec. II. An essence of both the MCF and (generalized) VCF approaches is shown to lie in an appropriate decomposition of the resolvent. The usual band-matrix assumption (with $H_{mn} = 0$ for all m and n such that $|m - n| > t$) is then found redundant: An arbitrary $t < \infty$ matrix H is shown to admit the VCF factorization.

The universal $t < \infty$ VCF method fully preserves a conceptual simplicity of the $t = 1$ special case⁹ reviewed briefly in Appendix A. In particular, the close interrelation between the $t = 1$ analytic continued fractional convergence and an asymptotic smoothness of H (Appendix B) is extended to $t > 1$. In Sec. III, this leads to an explicit algebraic nonperturbative definition of the effective Hamiltonians H^{eff} .

In Sec. IV, we apply the general method to the particular bound state problem (1.1). In the first step (Sec. IV A), we recall the results of Ref. 10 and regularize the anharmonicity in Eq. (1.1) by its Fourier symmetrization. Physically, this corresponds to an introduction of some particular relativistic corrections. Methodically this enables us to run effi-

ciently through the basic technicalities. Then, an asymptotic degeneracy of the nonrelativistic limit is found to admit a straightforward removal: We reinterpret the recently developed asymptotic formulas for wave functions¹¹ as an algebraic ansatz, and demonstrate its suitability by means of the computer symbolic manipulations in REDUCE (Sec. IV B).

Section V is a summary.

II. GENERAL HAMILTONIANS AND THE VECTORIAL CONTINUED FRACTIONS

An introduction of a model-space projector $P = \sum_{n=0}^M |n\rangle \langle n|$ and a trivial partitioning of the Schrödinger equation

$$\begin{aligned} P(H - E)(P + Q)|\psi\rangle &= 0, \\ Q(H - E)(Q + P)|\psi\rangle &= 0, \quad Q = 1 - P, \end{aligned} \quad (2.1)$$

enable one to eliminate the out-of-the-model-space components of the wave functions,

$$Q|\psi\rangle = (E - QHQ)^{-1}QHP|\varphi\rangle, \quad |\varphi\rangle = P|\psi\rangle. \quad (2.2)$$

The rest of Eq. (2.1) acquires the form of an effective finite-dimensional equation

$$H^{\text{eff}}|\varphi\rangle = E|\varphi\rangle, \quad (2.3)$$

where⁸

$$H^{\text{eff}} = PHP + PHQ(E - QHQ)^{-1}QHP. \quad (2.4)$$

Without any loss of generality, we may now consider a factorization

$$H - E = UL, \quad (2.5)$$

where U is an upper triangular matrix or any regular matrix with the property $QU = QUQ$. Similarly, the factor matrix L must be such that $LQ = QLQ$. Then, we may rewrite Eq. (2.4) in an explicitly factorized form

$$\begin{aligned} H^{\text{eff}} - E &= PULP - PULQ(1/QUQ)QULP \\ &= PULP - PUQLP = PUPLP \end{aligned} \quad (2.6)$$

born by Eq. (2.5). This is our starting point.

For the simple tridiagonal Hamiltonian (A3) (cf. Appendix A), the factors PHQ and QHP contain just one non-zero matrix element. Within any model space, the difference between H and H^{eff} concerns also a single element [cf. (A4) and (A5)],

$$H^{\text{eff}} = \begin{pmatrix} A_0 & C_1 & & \\ & \ddots & & \\ & B_{M-2} & A_{M-1} & C_M \\ & & B_{M-1} & G_M \end{pmatrix}, \quad G_M = E + \frac{1}{F_M}. \quad (2.7)$$

Thus we may define H^{eff} by means of the analytic continued fractions.⁹ A similar prescription is to be derived now for an entirely general matrix H .

A motivation for such an effort is the following. In the realistic (e.g., many-body) systems with complicated Hamiltonians H and simple trial Lanczos state $|0\rangle$, a tridiagonalization of H achieved by means of the Lanczos prescription¹² (A1) will lead to the complicated basis states $|1\rangle, |2\rangle, \dots$. Vice versa, the product $H|0\rangle$ will contain a number of the basis states $|1\rangle, |2\rangle, \dots$ whenever we require their reasonable simplicity,¹³

$$H|0\rangle = |0\rangle A_0 + |1\rangle B_0^{(1)} + |2\rangle B_0^{(2)} + \dots \quad (2.8a)$$

Similarly, in a repetition of this procedure,

$$H|1\rangle = |0\rangle C_1^{(1)} + |1\rangle A_1 + |2\rangle B_1^{(1)} + |3\rangle B_1^{(2)} + \dots \quad (2.8b)$$

a number of the new states will be infinite in principle. In this way, a tridiagonality of H will be lost and we may return, at least partially, to a free choice of the suitable basis states. At most, we may expect that the coefficients $B_k^{(j)}$ will be small for $j \gg 1$.

Formally, we may introduce the vectorial notation

$$(B_k^{(1)}, B_k^{(2)}, \dots) = \mathbf{B}_k^T, \quad \mathbf{B}_k = \begin{pmatrix} B_k^{(1)} \\ B_k^{(2)} \\ \vdots \end{pmatrix}, \quad |\mathbf{k}\rangle = \begin{pmatrix} |k\rangle \\ |k+1\rangle \\ \vdots \end{pmatrix},$$

generalize Eq. (A1),

$$H|\mathbf{k}\rangle = |0\rangle C_1^{(k)} + |1\rangle C_2^{(k-1)} + \dots + |k-1\rangle C_k^{(1)} + |\mathbf{k}\rangle A_k + |\mathbf{k}+1\rangle \cdot \mathbf{B}_k, \quad k = 0, 1, \dots, \quad (2.9)$$

and reinterpret our operator or general matrix H as tridiagonal in a purely formal "vectorial" partitioning

$$H = \begin{pmatrix} A_0 & C_1^T & & \\ \mathbf{B}_0 & A_1 & C_2^T & \\ & \mathbf{B}_1 & A_2 & C_3^T \\ & & \ddots & \ddots \end{pmatrix}. \quad (2.10)$$

When we introduce also the tilded diagonal matrices

$$\tilde{F}_k = \begin{pmatrix} F_k & & \\ & F_{k+1} & \\ & & \ddots \end{pmatrix},$$

auxiliary vectors $\mathbf{U}_k^T = (U_k^{(1)}, U_k^{(2)}, \dots)$, $\mathbf{L}_{k-1}^T = (L_{k-1}^{(1)}, L_{k-1}^{(2)}, \dots)$, $k = 1, 2, \dots$, and vectors with superscripts [omit-

ted components, $C_k^{T[m]} = (C_k^{(m+1)}, C_k^{(m+2)}, \dots)$, $k, m \geq 1$], we may write (2.5),

$$H - E = \begin{pmatrix} A_0 - E & C_1^T & & \\ \mathbf{B}_0 & A_1 - E & C_2^T & \\ & & \ddots & \ddots \end{pmatrix} = \begin{pmatrix} 1 & \mathbf{U}_1^T \tilde{F}_1 & & \\ & \mathbf{U}_2^T \tilde{F}_2 & & \\ & & \ddots & \\ & & & \tilde{F}_0^{-1} \end{pmatrix} \begin{pmatrix} 1 & & & \\ \tilde{F}_1 \cdot \mathbf{L}_0 & 1 & & \\ & \tilde{F}_2 \cdot \mathbf{L}_1 & 1 & \\ & & \ddots & \ddots \end{pmatrix}. \quad (2.11)$$

This prescription becomes an algebraic identity whenever we satisfy the relations

$$C_{k+1}^{(l)} = U_{k+1}^{(l)} + \sum_{j=1}^{\infty} U_{k+1}^{(l+j)} F_{k+l+j} L_{k+l}^{(j)}, \quad (2.12)$$

$$B_k^{(l)} = L_k^{(l)} + \sum_{j=1}^{\infty} U_{k+l+1}^{(j)} F_{k+l+j} L_k^{(l+j)}, \quad l = 1, 2, \dots,$$

and the $l = 0$ requirements

$$A_k - E = \frac{1}{F_k} + \sum_{j=1}^{\infty} U_{k+1}^{(j)} F_{k+j} L_k^{(j)}, \quad k = 0, 1, \dots \quad (2.13)$$

In the abbreviated notation, we may also write Eq. (2.12) as a recurrent definition of U 's and L 's,

$$U_{k+1}^{(l)} = C_{k+1}^{(l)} - \mathbf{U}_{k+1}^{T[l]} \cdot \tilde{F}_{k+l+1} \cdot \mathbf{L}_{k+l}^{[l]}, \quad (2.14)$$

$$L_k^{(l)} = B_k^{(l)} - \mathbf{U}_{k+l+1}^T \cdot \tilde{F}_{k+l+1} \cdot \mathbf{L}_k^{[l]}, \quad l = \dots, 2, 1.$$

Then, Eq. (2.13) will represent just a vectorial generalization of the continued fractions (A5),

$$1/F_k = A_k - E - \mathbf{U}_{k+1}^T \cdot \tilde{F}_{k+1} \cdot \mathbf{L}_k, \quad k = \dots, 2, 1, 0. \quad (2.15)$$

Thus, in analogy with (A9), the eigenvalue condition $\det(H - E) = 0$ acquires the simple VCF form $1/F_0 = 0$, i.e.,

$$E = A_0 - \mathbf{U}_1^T \cdot \tilde{F}_1 \cdot \mathbf{L}_0. \quad (2.16)$$

In the original Schrödinger equation, the factorized operator $H - E$ or $H^{\text{eff}} - E$ may be divided by its regular QUQ part. This leads to the generalized Eq. (A10),

$$\begin{pmatrix} 1/F_0 & & & \\ L_0^{(1)} & 1/F_1 & & \\ L_0^{(2)} & L_1^{(1)} & 1/F_2 & \\ & & \ddots & \ddots \end{pmatrix} \begin{pmatrix} \langle 0|\psi \rangle \\ \langle 1|\psi \rangle \\ \langle 2|\psi \rangle \\ \vdots \end{pmatrix} = 0 \quad (2.17)$$

with the numbers B_k replaced by the VCF vectors $\mathbf{L}_k$. The triangularity of the new matrix simplifies also the wave functions in a way paralleling Eq. (A.11),

$$\langle k|\psi \rangle = (-1)^k \cdot \langle 0|\psi \rangle \cdot \det \begin{pmatrix} F_1 \cdot L_0^{(1)} & 1 & 0 & \dots & 0 \\ F_2 \cdot L_0^{(2)} & F_2 \cdot L_1^{(1)} & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ F_k \cdot L_0^{(k)} & F_k \cdot L_1^{(k-1)} & \dots & F_k \cdot L_{k-1}^{(1)} \end{pmatrix}, \quad k = 1, 2, \dots \quad (2.18)$$

The VCF construction of bound states is completed.

III. AN ALGEBRAIC TRACTABILITY OF THE SPECIAL, ASYMPTOTICALLY SMOOTH MATRICES H

On an arbitrary fixed level of precision, we may always use truncations of the type

$$\begin{aligned} B_i^T &\approx (B_i^{(1)}, \dots, B_i^{(t)}, 0, 0, \dots), \\ C_j^T &\approx (C_j^{(1)}, \dots, C_j^{(s)}, 0, 0, \dots), \end{aligned} \quad (3.1)$$

since the scalar products $C_i^T \cdot B_j$ (contributing to the matrix elements of H^2) must converge. Moreover, in the spirit of Appendix B, we shall assume that

$$A_k \approx A_{k+1}, \quad B_k = C_{k+1}^* \approx B_{k+1} = C_{k+2}^*, \quad k \gg k_0, \quad (3.2)$$

i.e., $t_i = s_i = t \gg 1$ in (3.1). These restrictions still specify a sufficiently broad class of Hamiltonians.

A. The elementary factors

Let us recall (2.11) and interpret each factor as a product of t two-diagonal matrices. For the sake of clarity, we may also write

$$\begin{aligned} H - E &= L^+ DL, \\ L &= L_1 D_1 L_2 D_2 \cdots L_t D_t, \quad t \leq \infty, \end{aligned} \quad (3.3)$$

where $D = D^+$ and D_i are diagonal, $(L_i)_{mm} = 1$, $(L_i)_{m+1m} \neq 0$, and $(L_i)_{mn} = 0$ otherwise, $i = 1, 2, \dots, t$. The condition (3.2) may be combined with (3.3) in various ways. Here, we shall postulate that L_i are asymptotically constant and write

$$\begin{aligned} Q(H - E)Q &\approx QD_i^* \begin{pmatrix} 1 & d_i^* & & \\ & 1 & d_i^* & \\ & & \ddots & \ddots \end{pmatrix} \cdot D_{i-1}^* \begin{pmatrix} 1 & d_{i-1}^* & & \\ & 1 & d_{i-1}^* & \\ & & \ddots & \ddots \end{pmatrix} \cdot D_{i-2}^* \times \cdots \\ &\times \begin{pmatrix} 1 & d_1^* & & \\ & 1 & d_1^* & \\ & & \ddots & \ddots \end{pmatrix} \cdot D \begin{pmatrix} 1 & & & \\ d_1 & 1 & & \\ & d_1 & 1 & \\ & & \ddots & \ddots \end{pmatrix} \cdot D_1 \times \cdots \times \begin{pmatrix} 1 & & & \\ d_t & 1 & & \\ & d_t & 1 & \\ & & \ddots & \ddots \end{pmatrix} \cdot D_t \cdot Q. \end{aligned} \quad (3.4)$$

After a partial ordering of the (complex) parameters d_i ,

$$|d_i| \geq |d_{i+1}|,$$

we may also set $d_i = 0$ beyond some fixed and finite index $t = i_{\max}$ (on any predetermined level of approximation). Moreover, our "far-off-diagonal" truncation may be combined also with another approximation: $QD_1 = \text{const}, \dots, QD_{t-1} = \text{const}$ and, possibly, $QD = \text{const}$. In such a case, we obtain a new representation of the operator $Q(H - E)Q$,

$$Q(H - E)Q \approx QD_i^* \begin{pmatrix} e^{-\alpha_i^*} & e^{\alpha_i^*} & & \\ & e^{-\alpha_i^*} & e^{\alpha_i^*} & \\ & & \ddots & \ddots \end{pmatrix} \begin{pmatrix} e^{-\alpha_{i-1}^*} & e^{\alpha_{i-1}^*} & & \\ & e^{-\alpha_{i-1}^*} & e^{\alpha_{i-1}^*} & \\ & & \ddots & \ddots \end{pmatrix} \times \cdots \times \begin{pmatrix} e^{-\alpha_1} & e^{\alpha_1} & & \\ & e^{-\alpha_1} & e^{\alpha_1} & \\ & & \ddots & \ddots \end{pmatrix} D_t Q \quad (3.5)$$

after a reparametrization of $d_i = \exp 2\alpha_i$. Here, all the dependence on indices is to be carried by the diagonal outer factors D_i and D_i^+ .

In the simplest $t = 1$ special case, we get the quasiconstant behavior of the Hamiltonian in its asymptotic tridiagonal part,

$$Q(H - E)Q \approx QD_i^* \begin{pmatrix} \cdots & & & & \\ \cdots & 0 & e^{2i\text{Im}\alpha} & 2 \text{ch } 2 \text{Re } \alpha & e^{-2i\text{Im}\alpha} & 0 & \cdots \\ \cdots & & & & & & \end{pmatrix} D_i Q. \quad (3.6)$$

In particular, we get the real and symmetric Hamiltonians

$$D_i^{*-1} Q(H - E) Q D_i^{-1} \approx \begin{pmatrix} \cdots & & & & \\ \cdots & 1 & 2 \text{ch } 2\alpha & 1 & \cdots \\ & & \ddots & & \end{pmatrix} = \begin{pmatrix} \cdots & & & \\ \cdots & 121 & \cdots \\ & \ddots & \end{pmatrix} + 4 \text{sh}^2 \alpha \cdot I \quad (3.7)$$

for the real α 's. Similarly, we get the real asymptotics with $t = 2$,

$$H_{mn} - E\delta_{mn} \approx (D_2^*)_{mm} \left[\binom{4}{2+m-n} + 4(\text{sh}^2 \alpha_1 + \text{sh}^2 \alpha_2) \binom{2}{1+m-n} + 16 \text{sh}^2 \alpha_1 \text{sh}^2 \alpha_2 \delta_{mn} \right] (D_2)_{nn}, \quad m, n \gg M \gg 1, \quad (3.8)$$

etc. The general formulas may be found elsewhere.¹⁴

The ambiguity or symmetry between α_i and $-\alpha_i$ (or $\alpha_i \rightarrow -\alpha_i^*$, in general) will be removed later; in brief, it will be shown to intertwine the physical and unphysical wave functions. Thus, in the physical case, we obtain the unique prescriptions from Eq. (2.6). In particular, with $t = 1$, we get the effective matrix element

$$H_{MM}^{\text{eff}} - E \approx (D_1)_{MM}^2 \exp(-2\alpha), \quad M \gg 1 \quad (3.9a)$$

from (3.7) or, in general,

$$H_{MM}^{\text{eff}} - E \approx |D_1|_{MM}^2 \exp(-2 \operatorname{Re} \alpha), \quad M \gg 1 \quad (3.9b)$$

from (3.6). Similarly, with $t = 2$, we get

$$D_2^{-1}(H^{\text{eff}} - E)D_2^{-1} \approx PD_2^{-1}(H - E)D_2^{-1}P$$

$$= \begin{pmatrix} 0 & \cdots & 0 \\ \cdots & \exp 2(\alpha_1 + \alpha_2), & 2 \exp(\alpha_1 + \alpha_2) \operatorname{ch}(\alpha_1 - \alpha_2) \\ 0 \cdots 0, & 2 \exp(\alpha_1 + \alpha_2) \operatorname{ch}(\alpha_1 - \alpha_2), & 4 \operatorname{ch}^2(\alpha_1 - \alpha_2) + \exp 2(\alpha_1 + \alpha_2) \end{pmatrix}$$

$$\approx \begin{pmatrix} \cdots & 4 \operatorname{ch}^2(\alpha_1 - \alpha_2) + \exp[-2(\alpha_1 + \alpha_2)], & 2 \exp[-(\alpha_1 + \alpha_2)] \operatorname{ch}(\alpha_1 - \alpha_2) \\ \cdots & 2 \exp[-(\alpha_1 + \alpha_2)] \operatorname{ch}(\alpha_1 - \alpha_2), & \exp[-2(\alpha_1 + \alpha_2)] \end{pmatrix} \quad (3.10)$$

from (3.8), etc.

B. The difference-equation technique

The Q -projected part of the Schrödinger equation (2.1) may be rewritten in the form factorized in accord with Eq. (2.5),

$$QUQQLQ|\psi\rangle + QUQQLP|\psi\rangle = 0. \quad (3.11)$$

Obviously, the regular factor QUQ is redundant here. Moreover, we may ignore the first t rows ($t < \infty$) and write

$$\tilde{Q}LQ|\psi\rangle = 0, \quad \tilde{Q} = \sum_{n=M+t+1}^{\infty} |n\rangle\langle n|, \quad (3.12)$$

since $\tilde{Q}LP \equiv 0$. Now, the $(t+1)$ -term recurrences (3.12) are to be treated as a difference equation of the t th order.¹⁵

A complete factorization (3.3) of L converts Eq. (3.12) into the asymptotic relations

$$\tilde{Q}DL_1D_1 \cdots L_tD_tQ \begin{pmatrix} \langle M+1|\psi\rangle \\ \langle M+2|\psi\rangle \\ \vdots \end{pmatrix} = 0. \quad (3.13)$$

After the simplification (3.5), the latter relations become solvable in a closed form,

$$\langle M|\psi\rangle = \sum_{i=1}^t \lambda_i \psi_m^{[i]}, \quad \psi_m^{[i]} \approx (-d_i)^m, \quad m > M \gg 1. \quad (3.14)$$

For all H with t different values of d_i such that $|d_i| \neq 1$, the t independent solutions $\psi_m^{[i]}$ of our asymptotic Schrödinger equation (3.12) contain still the above-mentioned $d_i \rightarrow 1/d_i$ ambiguity. The standard requirement of existence of $\psi(r)$ with the finite norm,

$$\sum_{m=0}^{\infty} \langle \psi|m\rangle \langle m|\psi\rangle < \infty \quad (3.15)$$

implies that we may remove the ambiguity via the sufficient condition of convergence in Eq. (3.15),

$$|d_i| < 1, \quad i = 1, 2, \dots, t. \quad (3.16)$$

The boundary of the physical region with some $|d_i| = 1$ must be investigated separately.

In the above discussion, our use of factorizations is re-

dundant. The k th row of the band-matrix Schrödinger equation

$$B_{k-t}^{(t)} \langle k-t|\psi\rangle + B_{k-t+1}^{(t-1)} \langle k-t+1|\psi\rangle + \cdots + C_{k+1}^{(t)} \langle k+t|\psi\rangle = 0 \quad (3.17)$$

may be directly understood as a difference equation of the $2t$ th order.¹⁵ Its $2t$ independent solutions $\psi_k^{[i]}$, $i = \pm 1, \pm 2, \dots, \pm t$ complement the set (3.14) simply via the replacement $d_i \rightarrow 1/d_i$ for the negative super/subscripts i . We may preserve the former notation and denote the Jost solutions¹⁶ [compatible with (3.15)] by the upper positive superscripts.

The physical parameters λ_i and E in (3.14) are to be determined from the first $M+t+1$ rows of the Schrödinger equation omitted in our asymptotic subsystem of Eqs. (3.12). Vice versa, a variation of the submatrix PHP leads to a change of these coefficients in general. Hence the last t rows of the effective Eq. (2.3),

$$B_{M-2t+j}^{(t)} \psi_{M-2t+j} + \cdots + B_{M-t}^{(j)} \psi_{M-t} + g_{j1} \psi_{M-t+1} + \cdots + g_{jt} \psi_M = 0, \quad j = 1, 2, \dots, t, \quad (3.18)$$

must be satisfied by each of the t independent Jost states $\psi_k^{[i]}$, $k > M-2t$, $i = 1, 2, \dots, t$. In a compact notation with the $(t \times t)$ -dimensional matrices $g = g_m$, $(b_{m-1})_{ij} = B_{M-2t+j}^{(t+i-j)} (M+1 = t_0 + mt, \text{ cf. Appendix C})$, and

$$x_{ij} = \psi_{M-2t+i}^{[j]}, \quad y_{ij} = \psi_{M-t+i}^{[j]}, \quad i, j = 1, 2, \dots, t,$$

we must satisfy, therefore, the matrix identity $b_{m-1}x + g_my = 0$, i.e.,

$$g_m = -b_{m-1}xy^{-1}. \quad (3.19)$$

This formula is very important: Our knowledge of a complete system of Jost solutions becomes equivalent to an explicit knowledge of the effective Hamiltonian H^{eff} . Equation (3.19) complements the previous MCF or VCF definition of g_m and is in fact the most natural generalization of Eq. (2.7) to all $t \geq 1$.

IV. SCHRÖDINGER EQUATION WITH POLYNOMIAL POTENTIALS

A. The semirelativistic $p-r$ symmetrization of the Hamiltonian

A semirelativistic form or extension¹⁰

$$(T + V)\psi = E\psi, \quad V = \sum_{i=0}^{m_1} g_i r^{2i+2},$$

$$T = \sum_{j=0}^{m_2} h_j p^{2j+2} = (c^4 + p^2 c^2)^{1/2} - c^2 + O(p/c)^{2m_2+4} \quad (4.1)$$

of the anharmonic oscillator equation (1.1) degenerates to the exactly solvable harmonic oscillator for $m_1 = m_2 = 0$. With $m_1 = m_2 = 2q$, it remains symmetric with respect to the Fourier transformation $p \leftrightarrow r$. Here, it may be used as an illustrative example of the asymptotically band-matrix Hamiltonian since, in accord with Ref. 10, the asymptotically dominant part of H may be given the form

$$(T + V - E)_{mn} \approx \binom{4q+2}{2q+1+2m-2n} m^{2q+1} \times \left(1 + O\left(\frac{1}{m}\right)\right), \quad m \gg 1. \quad (4.2)$$

For the sufficiently large indices m and n , we may write

$$(T + V - E)_{mn} \approx \binom{2q}{q+m-n} + \sum_{l=1}^q \rho_l^{(q)} \binom{2q-2l}{q-l+m-n}, \quad (4.3)$$

where

$$\rho_1^{(1)} = \frac{1}{3}, \quad \rho_1^{(2)} = 8, \quad \rho_2^{(2)} = 16/5, \quad (4.4)$$

$$\rho_1^{(3)} = 20, \quad \rho_2^{(3)} = 48, \quad \rho_3^{(3)} = 64/7,$$

etc.

In the first nontrivial example with $q = 1$ and recurrences (4.1),

$$\psi_{n-1} + \frac{10}{3}\psi_n + \psi_{n+1} = 0, \quad n \gg 1 \quad (4.5)$$

we may put $\psi_n^{(1)} \sim (-d)^n$ and obtain $d = \frac{1}{3}$ or $d = 3$. In this case, the auxiliary VCF quantities degenerate to the ordinary continued fractions, and the finite approximants with $F_{M+1} = 0$ may be given an explicit form,

$$F_{M+1-k} = d [(d^{2k} - 1)/(d^{2k+2} - 1)], \quad k = 0, 1, \dots \quad (4.6)$$

This converges to the value $d_1 = \min(d, d^{-1})$ that is always smaller than 1. We observe a compatibility of the variational truncation with our difference-equation requirement (3.16).

In a straightforward way, the same analysis may be repeated for $t = q > 1$ as well. For example, the $q = 2$ recurrences (4.1),

$$\psi_{n-2} + 12\psi_{n-1} + 25.2\psi_n + 12\psi_{n+1} + \psi_{n+2} = 0, \quad n \gg 1, \quad (4.7)$$

may be solved by the ansatz (3.14) which gives⁶

$$d_1 = 1/d_{-1} = 5d_2 = 5/d_{-2} = 5 - 2\sqrt{5} \approx 0.527 \quad (4.8)$$

in an elementary way.

For a general value of $q = t$, the VCF convergence becomes less transparent. A basic idea of its analysis¹⁶ may be visualized as follows. First, we recall the partitioned notation of Appendix C and definitions

$$a_k - E = h_k s_k + l_{k+1} u_{k+1}, \quad b_k = h_{k+1} u_{k+1},$$

$$c_{k+1} = l_{k+1} s_{k+1}$$

of the $(t \times t)$ -dimensional matrices entering the VCF/MCF mapping $g_{k+1} (= h_{k+1} s_{k+1}) \rightarrow g_k$ (C5). By means of an ansatz

$$g_k = h_k s_k + l_{k+1} p_k u_{k+1}, \quad (4.9)$$

we convert these recurrences into an equivalent prescription

$$p_k = \gamma_1 p_{k+1} / (1 + \rho p_{k+1}) \gamma_2,$$

$$\gamma_1 = h_{k+1}^{-1} l_{k+2}, \quad \gamma_2 = u_{k+2} / s_{k+1}, \quad \rho = \gamma_2 \gamma_1. \quad (4.10)$$

Now, assuming that the k dependence of all the matrices is sufficiently weak here, the VCF/MCF convergence may be reinterpreted as a convergence of iterations of the mapping (4.10) performed at a fixed index $k \gg 1$. We may employ a spectral representation of the auxiliary matrices

$$\gamma_1 = \sum_{\epsilon_1 \in \Theta_1} |\epsilon_1\rangle \epsilon_1 \langle \epsilon_1|, \quad \gamma_2 = \sum_{\epsilon_2 \in \Theta_2} |\epsilon_2\rangle \epsilon_2 \langle \epsilon_2|, \quad (4.11)$$

and denote their maximal eigenvalues by a zero superscript. Immediately, we may notice then that the repeated multiplication of p 's by γ 's in (4.10) suppresses all the components not corresponding to $\epsilon_i = \epsilon_i^{(0)}$, $i = 1, 2$.

For the simplest case with the nondegenerate spectra Θ_i and unique values of $\epsilon_i^{(0)}$, $i = 1, 2$, a sufficient number of iterations in (4.10) converts the dominant component of p_k into a separable expression,

$$p_k = |\epsilon_1^{(0)}\rangle \kappa_k \langle \epsilon_2^{(0)}| + \text{corrections}. \quad (4.12)$$

Vice versa, an insertion of (4.12) changes the VCF/MCF mapping $g_{k+1} \rightarrow g_k$ or $p_{k+1} \rightarrow p_k$ into an approximately one-dimensional mapping $\kappa_{k+1} \rightarrow \kappa_k$,

$$\kappa_k = \epsilon_1^{(0)} \epsilon_2^{(0)} \kappa_{k+1} / (1 + \epsilon_1^{(0)} \epsilon_2^{(0)} \langle \epsilon_2^{(0)} | \epsilon_1^{(0)} \rangle \kappa_{k+1}) \quad (4.13)$$

with an easy analysis of convergence.

B. Effective Hamiltonians in the nonrelativistic case

The nonrelativistic $t = 2$ analog (1.1) of Eq. (4.7),

$$\psi_{n-2} + 4\psi_{n-1} + 6\psi_n + 4\psi_{n+1} + \psi_{n+2} = 0 \quad (4.14)$$

leads immediately to a quadruple leading-order asymptotic degeneracy⁷ $d_i = 1$, $i = 1-4$ of the Jost solutions (3.14). This result is easily generalized to any t . Indeed, from the initial values

$$A_{M+1} = A_{M+2} = \dots = 0,$$

$$B_M = B_{M+1} = \dots = C_{M+1} = \dots = 0, \quad (4.15)$$

$$B_{M-1}^{[1]} = B_{M-2}^{[2]} = \dots = C_M^{[1]} = \dots = 0, \quad M \gg 1$$

we obtain, after a reasonable amount of the algebraic manipulations, the sequences

$$F_{M+1-k} = \frac{k(k+1)\cdots(k+t-1)(1+O(k/M))}{(k+t)(k+t+1)\cdots(k+2t-1)},$$

$$U_{M+1-k}^{(j)} = \left(\frac{t}{j} \right) \frac{(k+t+j)(k+t+j+1)\cdots(k+2t-1)}{(k+j)(k+j+1)\cdots(k+t-1)} \times \left(1 + O\left(\frac{k}{M}\right) \right), \quad j=1,2,\dots,t, \quad k=1,2,\dots, \quad (4.16)$$

i.e., a rigorous solution of the VCF recurrences (2.14) and (2.15) in the asymptotic region.

In contrast to the $p \leftrightarrow r$ symmetric equation (4.1) with

$$|\langle n+1|\psi\rangle/\langle n|\psi\rangle| \approx \max_i |d_i| < 1, \quad n \gg 1, \quad (4.17)$$

our present degeneracy of asymptotics need not be considered as a shortcoming. In fact, we may write

$$\langle n|\psi\rangle = (-1)^n \exp \xi_n, \quad (4.18)$$

where ξ_n should be some asymptotically smooth function of the index n . Its important merit lies in a possibility of inserting the Taylor series

$$\begin{aligned} \exp \xi_{n+k} &= \exp \xi_n + k \frac{d}{dn} \exp \xi_n + \cdots \\ &= \exp \xi_n (1 + k \xi'_n + \frac{1}{2} k^2 (\xi''_n + \xi'^2_n) + \cdots) \end{aligned} \quad (4.19)$$

in the left-hand side $V|\psi\rangle$ of equations of the type (4.14), $QV|\psi\rangle = 0$. Then, whenever we replace the right-hand side zero by the leading-order estimate of the kinetic energy contribution $-T|\psi\rangle \approx 4\psi_n$, $n \gg 1$, we obtain an estimate of ξ'_n or

$$\xi'_n \approx \text{const} \times n^{(1-t)/2t}, \quad n \gg 1, \quad (4.20)$$

in accord with Ref. 11.

In the next step, we may notice that all the corrections in (4.19) are given as derivatives of ξ_n . An inclusion of the precise matrix elements of H may only modify and convert Eq. (4.20) into a series

$$\xi'_n = \sum_{m=1}^L \beta_m \rho^m + O(\rho^{L+1}), \quad \rho = n^{-1/2t} \ll 1, \quad (4.21)$$

where, in accord with Eq. (4.20),

$$\beta_1 = \cdots = \beta_{t-2} = 0, \quad \beta_{t-1} \neq 0. \quad (4.22)$$

In analogy with the Jost solutions pertaining to the Hamiltonians with the nondegenerate parametrization (3.5), our present expansion (4.21) may also be used as an ansatz. It transforms the asymptotic anharmonic oscillator Schrödinger equation [say, Eq. (3.17) with $k = n \gg 1$] into a power-series requirement of the implicit general form

$$\sum_{k=0}^{\infty} \rho^k R_k(E, l, g_1, \dots, g_t, \beta_1, \beta_2, \dots) = 0. \quad (4.23)$$

Due to a linear independence of the powers of ρ , the conditions

$$R_k = 0, \quad k = 0, 1, \dots, \quad (4.24)$$

are to be solved as a nonlinear algebraic set of coupled equations for the coefficients β_i in (4.21) or (4.18) and (4.19).

1. $t=2$

Let us start our analysis from the simplest nontrivial quartic oscillator problem (1.1) with $t=2$. Performing the algebraic construction of the functions R_k in (4.24) on the computer (in REDUCE), we find that the first four items are identically zeros. This reflects the asymptotically correct behavior of our ansatz (4.18). From $R_4 = 0$ we obtain

$$\beta_1^{(4)} = -4/g_2. \quad (4.25)$$

In full analogy with our preceding discussion, the four independent complex roots $\beta_1^{(j)}$ of this equation with $j = \pm 1, \pm 2$ comprise also the physical Jost coefficients

$$\beta_1^{(1,2)} = (-1 \pm i) \cdot |g_2^{-1/4}|. \quad (4.26)$$

Similarly, the further items of Eq. (4.24) remain linear in the unknown coefficients and determine the respective sequence of their values

$$\begin{aligned} \beta_2 &= 0, \\ \beta_3 &= \frac{1}{8}(1 \pm i)(\frac{1}{2} - g_1)|g_2^{-3/4}|, \\ \beta_4 &= -\frac{1}{8}, \\ \beta_5 &= \frac{1}{16}(1 \mp i)[(E + 2l + 3)g_2 \\ &\quad + \frac{1}{8}(g_1 + 1)^2 - \frac{1}{2}] \cdot |g_2^{-5/4}|, \\ \beta_6 &= (\mp i/32)(3 - g_1) \cdot |g_2^{-1/2}|, \\ &\vdots \end{aligned} \quad (4.27)$$

Up to a misprint in β_5 and missing β_6 , they agree with the $g_2 = 1$ results of Ref. 11, i.e., with the wave function asymptotics

$$\begin{aligned} \psi_n^{(j)} &= (-1)^n n^{-5/8} \exp(\frac{4}{3} \beta_1^{(j)} n^{3/4} + 4\beta_3^{(j)} n^{1/4}) \\ &\quad \times \exp(-4\beta_5^{(j)} n^{-1/4} - 2\beta_6^{(j)} n^{-1/2} \\ &\quad + O(n^{-3/4})), \quad j = 1, 2, \end{aligned} \quad (4.28)$$

tested and verified numerically for $g_1 = 1$. Of course, such a choice simplifies also the higher-order coefficients here,

$$\begin{aligned} \beta_6^{(j)} &= \frac{1}{32} \beta_1^{(j)2}, \\ \beta_7^{(j)} &= \frac{1}{32} \beta_1^{(j)3} (-4\alpha^2 + 5\alpha + \frac{1}{4}E + \frac{1}{12}), \quad \alpha = \frac{1}{2}l + \frac{1}{2}, \\ \beta_8 &= \frac{3}{8}\alpha - \frac{3}{32}E + \frac{1}{32}, \\ \beta_9^{(j)} &= \frac{1}{32} \beta_1^{(j)} (11\alpha^2 + \frac{1}{4}\alpha(10E - 21) \\ &\quad - \frac{1}{16}(3E^2 - 3E + \frac{83}{32})), \\ \beta_{10}^{(j)} &= \frac{1}{32} \beta_1^{(j)2} (-3\alpha^2 + \frac{3}{2}\alpha - \frac{3}{8}E + \frac{23}{64}), \\ &\vdots \end{aligned} \quad (4.29)$$

2. $t=3$

In the sextic anharmonic oscillator example (1.1) with $g_1 = 1$ (scaling of the scale) and $t=3$, we may proceed along the same lines as above. After confirming Eq. (4.22) as a consequence of (4.24) at $k \leq 6$, we obtain the first nontrivial algebraic equation $R_k = 0$ at $k = 12$,

$$\beta_2^{(j)6} = 4/g_3. \quad (4.30)$$

We may specify the Jost roots $\beta_2^{(j)}$, $j = 1, 2$, and 3, by the simple condition

$$\text{sgn Re } \beta_2^{(j)} = -1, \quad j = 1, 2, \dots, t. \quad (4.31)$$

In the forthcoming steps, we obtain the zero odd coefficients

$$\beta_3 = \beta_5 = \beta_7 = \beta_9 = 0,$$

and the nontrivial contributions

$$\begin{aligned}\beta_4^{(j)} &= \frac{1}{24} g_2 \beta_2^{(j)5}, \\ \beta_6^{(j)} &= \frac{1}{24} \left[-16 + (-1 + g_2^2/4g_3) \beta_2^{(j)3} \right], \\ \beta_8^{(j)} &= \frac{1}{24} \left[\frac{1}{6} g_2 \beta_2^{(j)4} \right. \\ &\quad \left. + \left(\frac{7}{216} \frac{g_2^3}{g_3} - \frac{1}{2} \frac{g_2}{g_3} - 8\alpha - E \right) \beta_2^{(j)} \right],\end{aligned}\quad (4.32)$$

to the physical wave functions

$$\begin{aligned}\langle n | \psi \rangle &= \sum_{j=1}^3 (-1)^n \lambda_j n^{\beta_j^{(j)}} \\ &\quad \times \exp \left(\frac{3}{2} \beta_2^{(j)} n^{2/3} + \frac{g_2}{2\beta_2^{(j)} g_3} n^{1/3} \right) \\ &\quad \times \exp(-3\beta_8^{(j)} n^{-1/3} + O(n^{-2/3})), \quad n \gg 1.\end{aligned}\quad (4.33)$$

3. $t=4$

A knowledge of the first few solutions with increasing t simplifies the manipulations in REDUCE—we may incorporate there immediately the relations (4.22), formula

$$\beta_{t-1}^{2t-1} = (-1)^{t+1} \cdot (4/g_t) \quad (4.34)$$

derived in the second paper of Ref. 11 and further hypotheses inspired by the similarities of the preceding solutions. We must be careful, of course. In particular, the octic oscillator analog

$$\begin{aligned}\psi_n^{(j)} &= (-1)^n n^{\beta_8^{(j)}} \exp \left(\frac{8}{3} \beta_3^{(j)} n^{5/8} + \frac{8}{3} \beta_5^{(j)} n^{3/8} \right. \\ &\quad \left. + 8\beta_7^{(j)} n^{1/8} \exp(-8\beta_9^{(j)} n^{-1/8} + O(n^{-1/4})), \right. \\ &\quad \left. n \gg 1, \quad t=4,\end{aligned}\quad (4.35)$$

of (4.33) or (4.28) shows that an assumption $\beta_8 = 0$ is not correct. In detail, Eq. (4.24) with $k=24, 26, 28, 29, 30$, etc. leads to the respective sets of coefficients

$$\begin{aligned}\beta_3^{(j)} &= (-4/g_4)^{1/8}, \\ \beta_5^{(j)} &= -\frac{1}{32} g_3 \beta_3^{(j)7}, \\ \beta_7^{(j)} &= \frac{1}{32} \beta_3^{(j)5} \left(g_2 - \frac{5}{16} \frac{g_3^2}{g_4} \right), \quad \beta_8^{(j)} = -\frac{11}{16}, \\ \beta_9^{(j)} &= \frac{1}{32} \beta_3^{(j)3} \left(-\frac{4}{3} + \frac{3g_2 g_3}{8g_4} - \frac{11g_3^3}{128g_4^2} \right), \\ &\vdots\end{aligned}\quad (4.36)$$

with the scaled coupling $g_1 = 1$ again. These formulas are very suitable for an analysis of the various kinds of limits, but this is beyond the scope of the present paper.

V. SUMMARY

One of the most efficient numerical approaches to polynomial interactions is known to be a diagonalization in the standard oscillator basis. In a quasiperturbative, less numerical setting, this technique has recently been shown equivalent to a systematic algebraic construction of the MCF fixed-point approximants. Here, we have succeeded in solving the corresponding nonlinear systems of algebraic equations in a non-numerical manner. Up to the octic anharmonic oscillators, the explicit asymptotic-series representations of the effective Hamiltonians have been obtained.

Methodically, the two aspects of the technique deserve special attention. First, a similarity in structure of the asymptotically degenerate and nondegenerate Hamiltonians has been recovered. Conceptually, this makes the whole method extremely simple. The present constructions illustrate also its purely pragmatic efficiency.

Second, the underlying factorization has been given a “final” form—its vectorially partitioned character does not necessitate any sparse-matrix assumption anymore. In this sense, we believe in its further methodical development, especially via the various straightforward extensions of the underlying ansatz.

APPENDIX A: TRIDIAGONAL HAMILTONIANS AS A METHODOLOGICAL GUIDE

At the very beginning of the standard Lanczos numerical tridiagonalization of H , we have to pick up an arbitrary trial vector $|0\rangle$. Then, by means of its repeated multiplication by the operator H , we may generate the basis,¹²

$$\begin{aligned}|1\rangle &= (1/B_0) [H|0\rangle - |0\rangle \cdot A_0], \\ &\vdots \\ |k+1\rangle &= (1/B_k) [H|k\rangle - |k\rangle \cdot A_k \\ &\quad - |k-1\rangle \cdot C_k], \quad k=1, 2, \dots\end{aligned}\quad (A1)$$

The natural condition of orthonormality

$$\langle m | n \rangle = \begin{cases} 0, & m \neq n, \\ 1, & m = n \geq 0, \end{cases}\quad (A2)$$

determines all the coefficients in (A1) uniquely.

The operator H becomes represented by the tridiagonal matrix

$$H = \begin{pmatrix} A_0 & C_1 & & \\ B_0 & A_1 & C_2 & \\ & B_1 & A_2 & C_3 \\ & & & \ddots \end{pmatrix}\quad (A3)$$

in the basis (A1). As a consequence, we may introduce the factorization (2.5),

$$H - E = \begin{pmatrix} 1 & C_1 F_1 & & \\ & 1 & C_2 F_2 & \\ & & \ddots & \\ & & & 1/F_1 & \ddots \end{pmatrix} \cdot \begin{pmatrix} 1/F_0 & & & \\ & 1/F_1 & & \\ & & \ddots & \\ & & & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & & & \\ F_1 B_0 & 1 & & \\ & F_2 B_1 & 1 & \\ & & \ddots & \ddots \end{pmatrix},\quad (A4)$$

where the auxiliary quantities F_k have only to satisfy the recurrences

$$1/F_k = A_k - E - C_{k+1}F_{k+1}B_k, \quad k \geq 0. \quad (\text{A5})$$

Formally, this enables us to rewrite the secular determinant in the factorized form

$$\det(H - E) = \left(\prod_{k=0}^{\infty} F_k \right)^{-1}. \quad (\text{A6})$$

In the computational practice,¹⁷ we employ usually the truncation of H ,

$$\begin{aligned} A_{N+1} &= A_{N+2} = \cdots = 0, \\ C_{N+1} &= \cdots = 0, \quad B_N = \cdots = 0, \quad N \ll 1. \end{aligned} \quad (\text{A7})$$

In the limit $N \rightarrow \infty$, this leads to the results equivalent to an exact solution. In the present setting, this simplifies also an interpretation of recurrences (A5)—we may use $F_{N+1} = 0$ as an initial value and identify the quantities F_k with the analytic continued fractions,¹⁸

$$F_k = (A_k - E - C_{k+1}B_k/(A_{k+1} - E - \cdots))^{-1}. \quad (\text{A8})$$

For an arbitrary finite cutoff parameter $N < \infty$, the Schrödinger equation (1.1) possesses a nontrivial solution if and only if $\det(H - E) = 0$. Here, we may compare Eqs. (A5) and (A6) and see that the binding energies will coincide with the roots of the analytic continued fractional “Green’s function” $1/F_0$,

$$E = A_0 - C_1F_1(E)B_0. \quad (\text{A9})$$

Moreover, we may also omit the regular factor from Eq. (1.1) and, noticing that the first row of the resulting new form of our Schrödinger equation

$$\begin{pmatrix} 1/F_0 & \cdots & \cdots \\ B_0 & 1/F_1 & \cdots \\ 0 & B_1 & 1/F_2 \\ \vdots & \vdots & \vdots \end{pmatrix} \begin{pmatrix} \langle 0|\psi \rangle \\ \langle 1|\psi \rangle \\ \langle 2|\psi \rangle \\ \vdots \end{pmatrix} = 0 \quad (\text{A10})$$

becomes satisfied identically [cf. Eq. (A9)], we obtain an explicit continued fractional formula

$$\begin{aligned} \langle k|\psi \rangle &= -F_k B_{k-1} \langle k-1|\psi \rangle \\ &= (-1)^k \prod_{m=0}^{k-1} B_m F_{m+1} \cdot \langle 0|\psi \rangle. \end{aligned} \quad (\text{A11})$$

This defines the bound states.

APPENDIX B: THE FIXED POINT EXPANSIONS

In practice, a continued-fractional convergence $N \rightarrow \infty$ is usually very quick. This may easily be understood as a consequence of the weakening k dependence of the mapping $F_{k+1} \rightarrow F_k$ (A5) for the increasing indices $k \gg 1$. Under such an assumption (summarizing in fact just the numerical experience), our knowledge of the large number of quantities $F_N, F_{N-1}, \dots, F_{k+1}, F_k$ becomes redundant. Indeed, a weak k dependence of F_k ’s for $k \gg 1$ implies that we may expect that $F_{k+1} \approx F_k$ may be approximated by a “fixed-point” root $F_k^{[0]}$ of the simple quadratic equation

$$1/F_k^{[0]} = A_k - E - C_{k+1}F_k^{[0]}B_k, \quad k \geq 0. \quad (\text{B1})$$

An ambiguity of this definition may easily be removed by

means of the stability criterion—the physical root of (B1) becomes unique.

In accord with Eq. (A9), the quantity $1/F_0$ (or, in general, $1/F_k$) may be interpreted as a component of the Feshbach effective Hamiltonian.^{8,9} In a perturbative spirit of Ref. 7, we may also study the corrections

$$F_k^{(1)} = F_k - F_k^{[0]}. \quad (\text{B2})$$

Indeed, when we rewrite Eq. (A5) in a new form

$$1/(F_k^{(1)} + F_k^{[0]}) = A_k - E - C_{k+1}(F_{k+1}^{(1)} + F_{k+1}^{[0]})B_k, \quad (\text{B3})$$

we may subtract the definition (B1) and eliminate, say, the parameters $A_k - E$,

$$\begin{aligned} (1/F_k^{[0]})F_k^{(1)} [1/(F_k^{(1)} + F_k^{[0]})] \\ = C_{k+1}(F_{k+1}^{(1)} + F_{k+1}^{[0]} - F_k^{[0]})B_k. \end{aligned}$$

This may be rewritten as a new rational mapping

$$F_k^{(1)} = (\alpha_k^{(1)} + \beta_k^{(1)}F_{k+1}^{(1)})/(\gamma_k^{(1)} - F_{k+1}^{(1)}), \quad k = 0, 1, \dots \quad (\text{B4})$$

with the k -dependent parameters

$$\begin{aligned} \alpha_k^{(1)} &= F_k^{[0]}(F_{k+1}^{[0]} - F_k^{[0]}), \quad \beta_k^{(1)} = F_k^{[0]}, \\ \gamma_k^{(1)} &= (B_k C_{k+1})^{-1}(A_k - E) - F_{k+1}^{[0]} \\ &= (F_k^{[0]}B_k C_{k+1})^{-1} + F_k^{[0]} - F_{k+1}^{[0]}. \end{aligned} \quad (\text{B5})$$

Obviously, the subtraction of the type (B2) may easily be iterated. First, we have to specify the higher-order fixed-point approximants $F_k^{[n]}$ as the roots of the generalized Eqs. (B1),

$$F_k^{[n]} = (\alpha_k^{(n)} + \beta_k^{(n)}F_{k+1}^{[n]})/(\gamma_k^{(n)} - F_{k+1}^{[n]}), \quad n \geq 1. \quad (\text{B6})$$

Next, we notice a uniqueness of the definition (B6): one of the roots represents just a return to the solution discarded in the zero-order equation (B1). Thus assuming that the iterations

$$F_k = F_k^{[0]} + F_k^{[1]} + \cdots + F_k^{[n]} + F_k^{(n+1)} \quad (\text{B7})$$

converge, the smallest roots of Eq. (B6) may be treated as “physical.”

In a way analogous to the derivation of Eq. (B4), we may rewrite Eq. (A5) in the equivalent form

$$\begin{aligned} F_k^{(n+1)} &= (\alpha_k^{(n+1)} + \beta_k^{(n+1)}F_{k+1}^{(n+1)})/(\gamma_k^{(n+1)} \\ &\quad - F_{k+1}^{(n+1)}), \quad k, n \geq 0. \end{aligned} \quad (\text{B8})$$

This completes our fixed-point (FP) construction of the new expansion (B7). Indeed, we obtain the recurrent definitions of the relevant coefficients

$$\begin{aligned} \gamma_k^{(n+1)} &= \gamma_k^{(n)} - F_{k+1}^{[n]} = \frac{A_k - E}{B_k C_{k+1}} - \sum_{m=0}^n F_{k+1}^{[m]}, \\ \alpha_k^{(n+1)} &= \alpha_k^{(n)} + \beta_k^{(n)}F_{k+1}^{[n]} - \gamma_k^{(n+1)}F_k^{[n]} \\ &= \sum_{m=0}^n \sum_{l=0}^n F_k^{[m]}F_{k+1}^{[l]} \\ &\quad + \frac{1}{B_k C_{k+1}} \left[1 - (A_k - E) \sum_{m=0}^n F_k^{[m]} \right], \\ \beta_k^{(n)} &= \beta_k^{(n)} + F_k^{[n]} = \sum_{m=0}^n F_k^{[m]} \end{aligned} \quad (\text{B9})$$

valid for any fixed FP order n and variable index k . In the large- n limit, we get

$$\beta_k^{(n)} \rightarrow F_k, \quad \gamma_k^{(n)} \rightarrow (B_k F_k C_{k+1})^{-1}, \quad \alpha_k^{(n)} \rightarrow 0. \quad (\text{B10})$$

These relations enable us to prove or analyze the convergence of the FP alternative (B7) to the continued fractional expansion (A5) in detail. Their generalization to $t > 1$ is straightforward. It may be found elsewhere.^{7,16}

APPENDIX C: THE VCF/MCF EQUIVALENCE FOR $t < \infty$

For the $(2t + 1)$ -diagonal matrices

$$H = \begin{pmatrix} A_0 & C_1^{(1)} & \cdots & C_1^{(t)} & 0 & \cdots & 0 \\ & & & \cdots & & & \\ 0 & \cdots & & 0 & B_{k-t}^{(t)} & \cdots & C_{k+1}^{(t)} & 0 & \cdots & 0 \\ & & & \cdots & & & & & & \\ 0 & \cdots & & 0 & B_{N-t}^{(t)} & B_{N-t+1}^{(t-1)} & \cdots & B_{N-1}^{(1)} & A_N \end{pmatrix}, \quad N \leq \infty, \quad (\text{C1})$$

a block-tridiagonal repartitioning may be introduced,

$$H = \begin{pmatrix} a_0 & c_1 & 0 & \cdots & 0 \\ b_0 & a_1 & c_2 & 0 & \cdots & 0 \\ & & \cdots & & & \\ 0 & \cdots & 0 & b_{m-1} & a_n \end{pmatrix}, \quad nt + t_0 = N + 1, \quad (\text{C2})$$

with the $(t_k \times t_k)$ -dimensional submatrices a_k and $t_0 \leq t_1 = t_2 = \cdots = t$. Then, the VCF factorization (2.11) may either be partitioned,

$$H - E = \begin{pmatrix} h_0 & l_1 & & \\ & h_1 & l_2 & \\ & & \ddots & \\ & & & h_n \end{pmatrix} \times \begin{pmatrix} s_0 & & & \\ u_1 & s_1 & & \\ & & \ddots & \\ & & & u_n & s_n \end{pmatrix}, \quad \dim h_k = t_k, \quad (\text{C3})$$

or replaced by the new form of the general decomposition (2.5), namely, by the formula

$$H - E = \begin{pmatrix} I & c_1 f_1 & & \\ & I & c_2 f_2 & \\ & & \ddots & \\ & & & I \end{pmatrix} \times \begin{pmatrix} 1/f_0 & & & \\ & 1/f_1 & & \\ & & \ddots & \\ & & & 1/f_n \end{pmatrix} \times \begin{pmatrix} I & & & \\ f_1 b_0 & I & & \\ & & \ddots & \\ & & & f_n b_{n-1} & I \end{pmatrix}, \quad n \leq \infty. \quad (\text{C4})$$

The related recurrences

$$1/f_k = a_k - EI - c_{k+1} f_{k+1} b_k, \quad k = n, n-1, \dots, 0, \quad (\text{C5})$$

define simply the $(t \times t)$ -dimensional generalization of the continued fractions⁵ in the limit $N \rightarrow \infty$.

A comparison of Eqs. (C3) and (C4) implies that

$$f_k = (h_k s_k)^{-1}, \quad k = n, n-1, \dots, 0, \quad (\text{C6})$$

$$b_{m-1} = h_m u_m, \quad c_m = l_m s_m, \quad m = 1, 2, \dots, n,$$

so that the MCF quantities become defined uniquely in terms of their VCF counterparts. Whenever we require that, e.g., $(s_m)_{ii} = 1$, the opposite is also true for a broad class of Hamiltonians.

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